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(54) **RESIZABLE DEVICE STAND**

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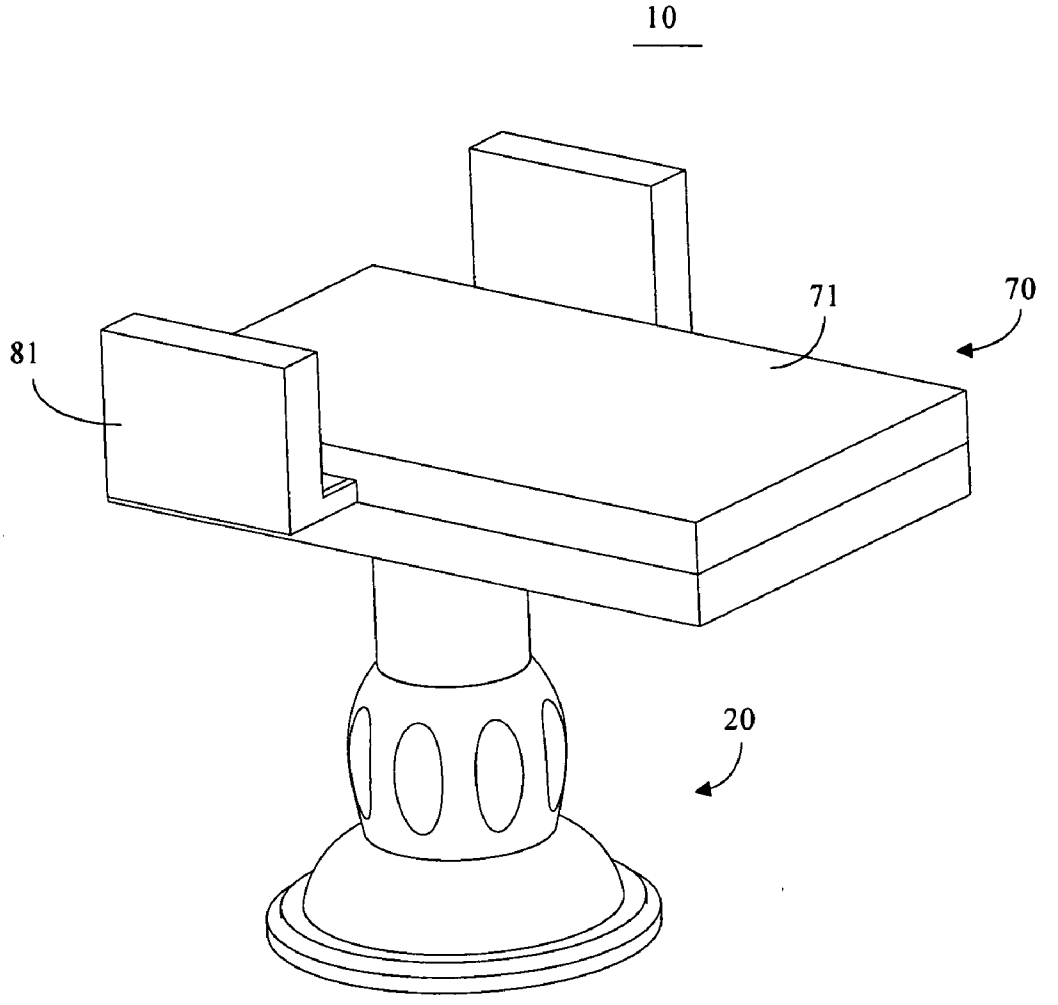
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(57) **ABSTRACT**

A stand includes a suction base and a load-supporting module mounted on the suction base. The load-supporting module includes a housing to house a pair of clamps, a gear, and a locking structure. The house defines a groove therein. Each clamp includes a vertical part and a horizontal part. The horizontal part includes a rack gear, and is received by the groove. The vertical parts are positioned upwardly from opposite sides of the housing respectively. The gear is rotatably mounted in the housing and meshes with the rack gears respectively. The locking structure is used for locking or unlocking the clamps. The stand is adapted for supporting different types and sizes of devices.



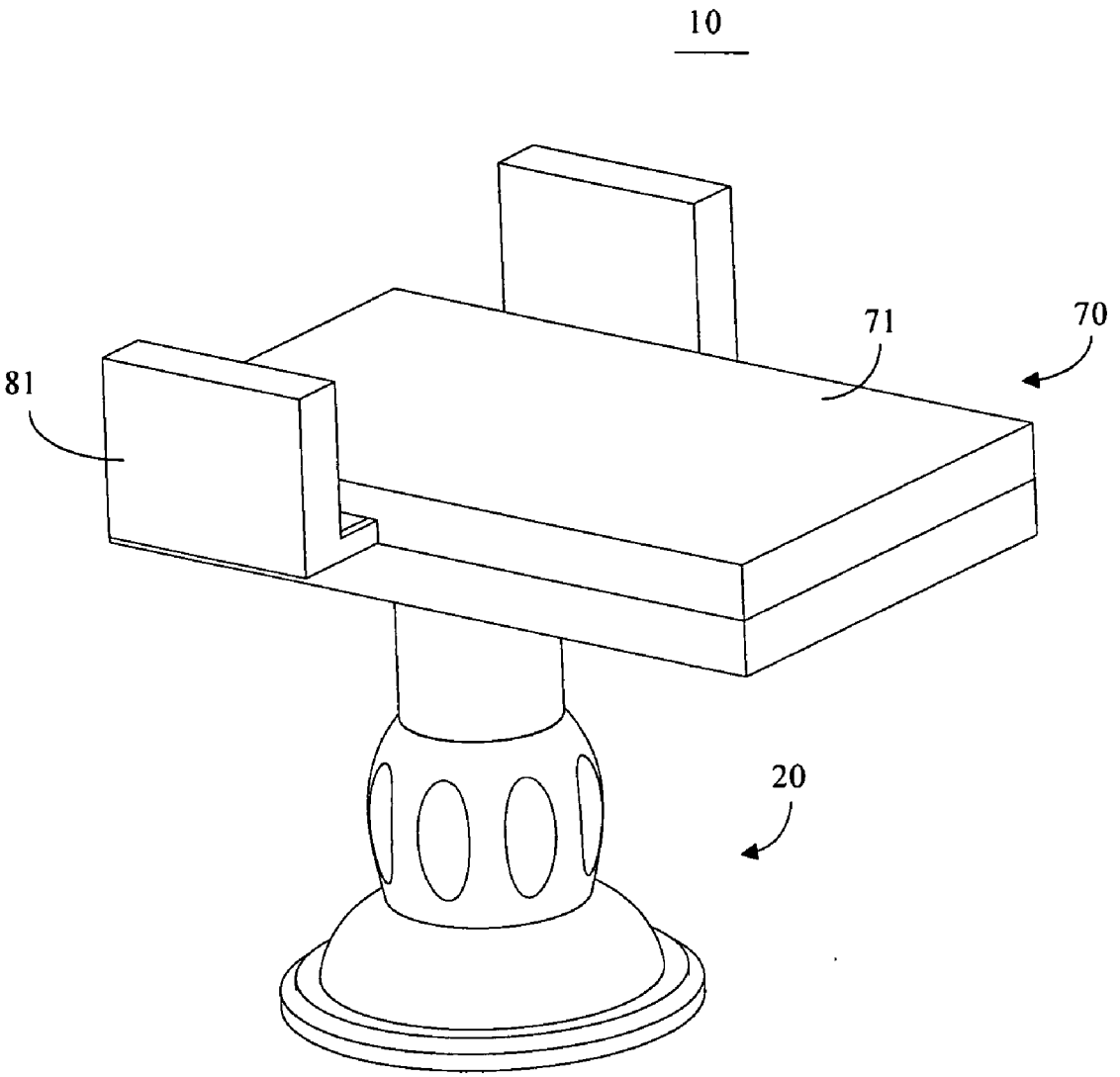


FIG. 1

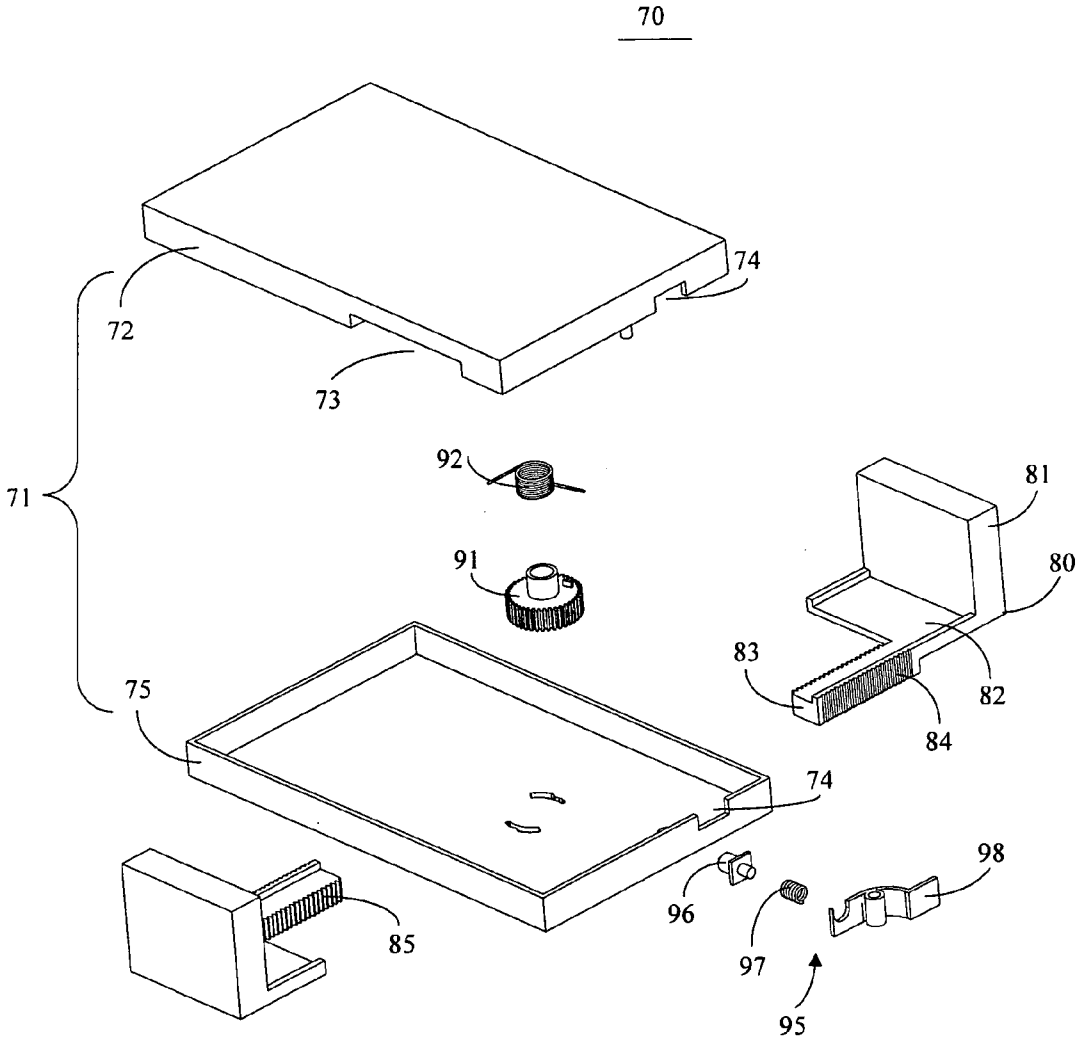


FIG. 2

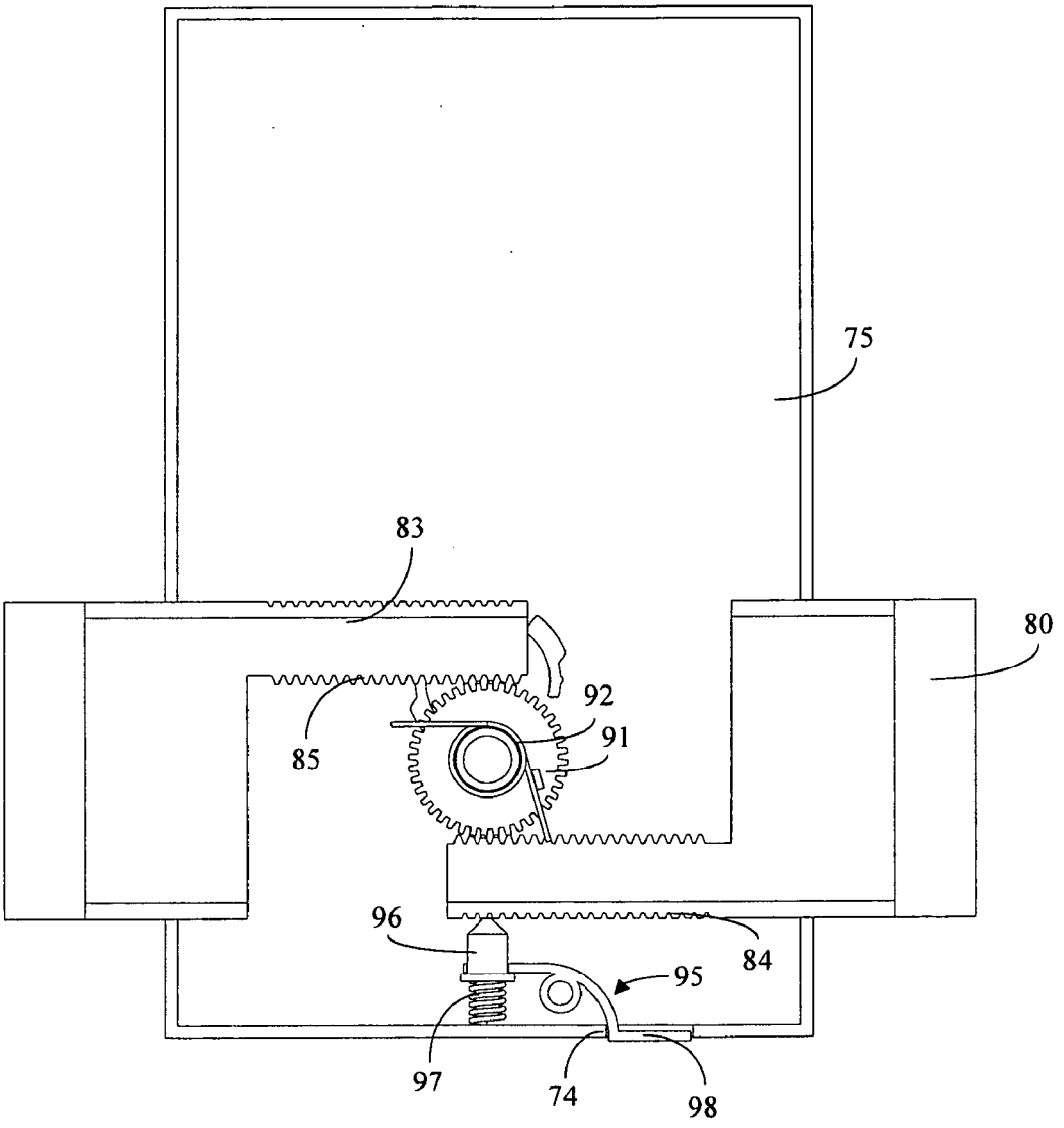


FIG. 3

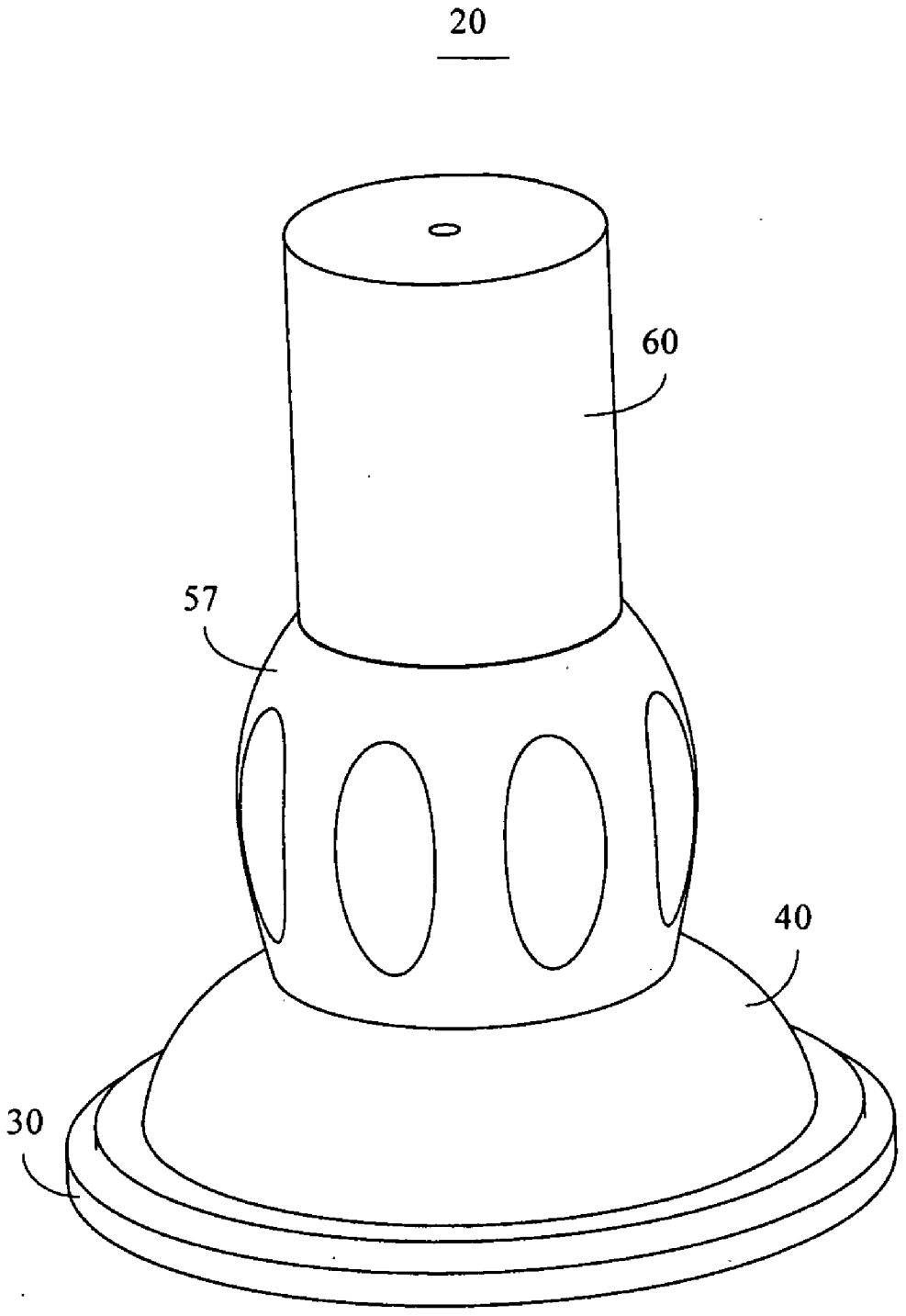


FIG. 4

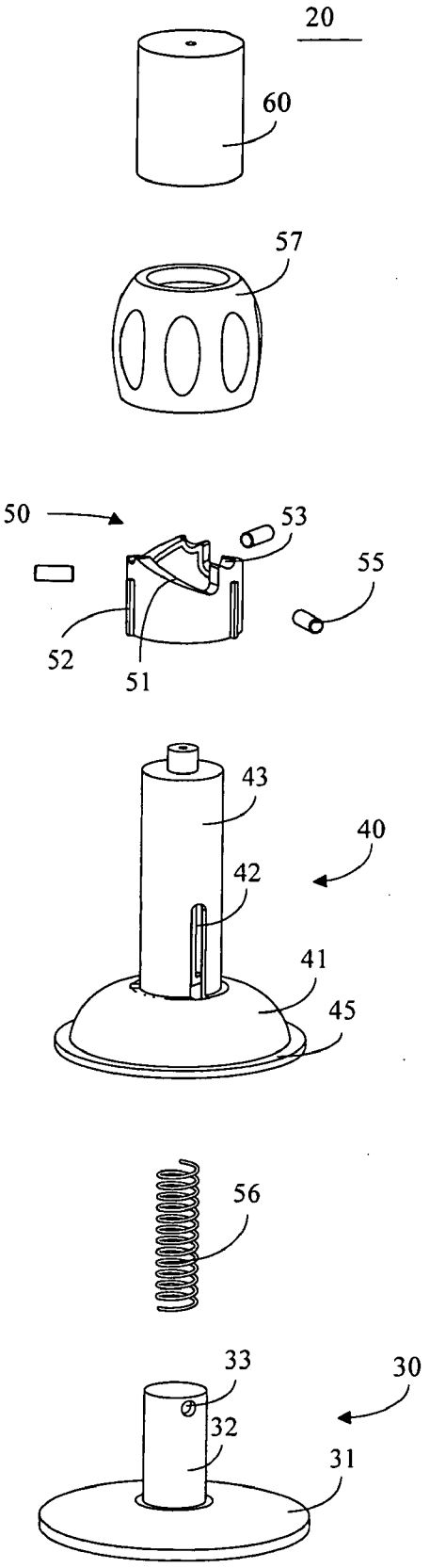


FIG. 5

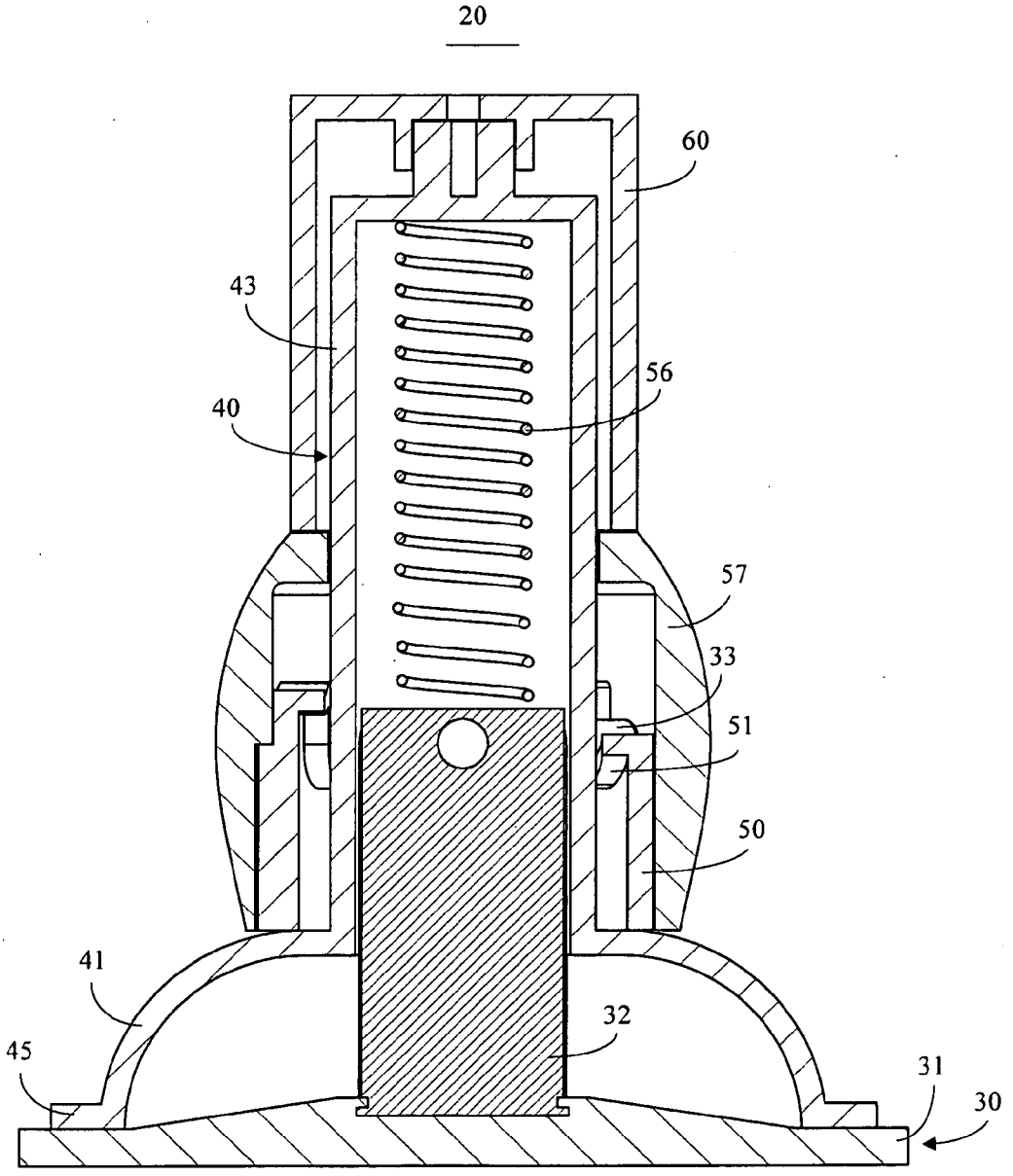


FIG. 6

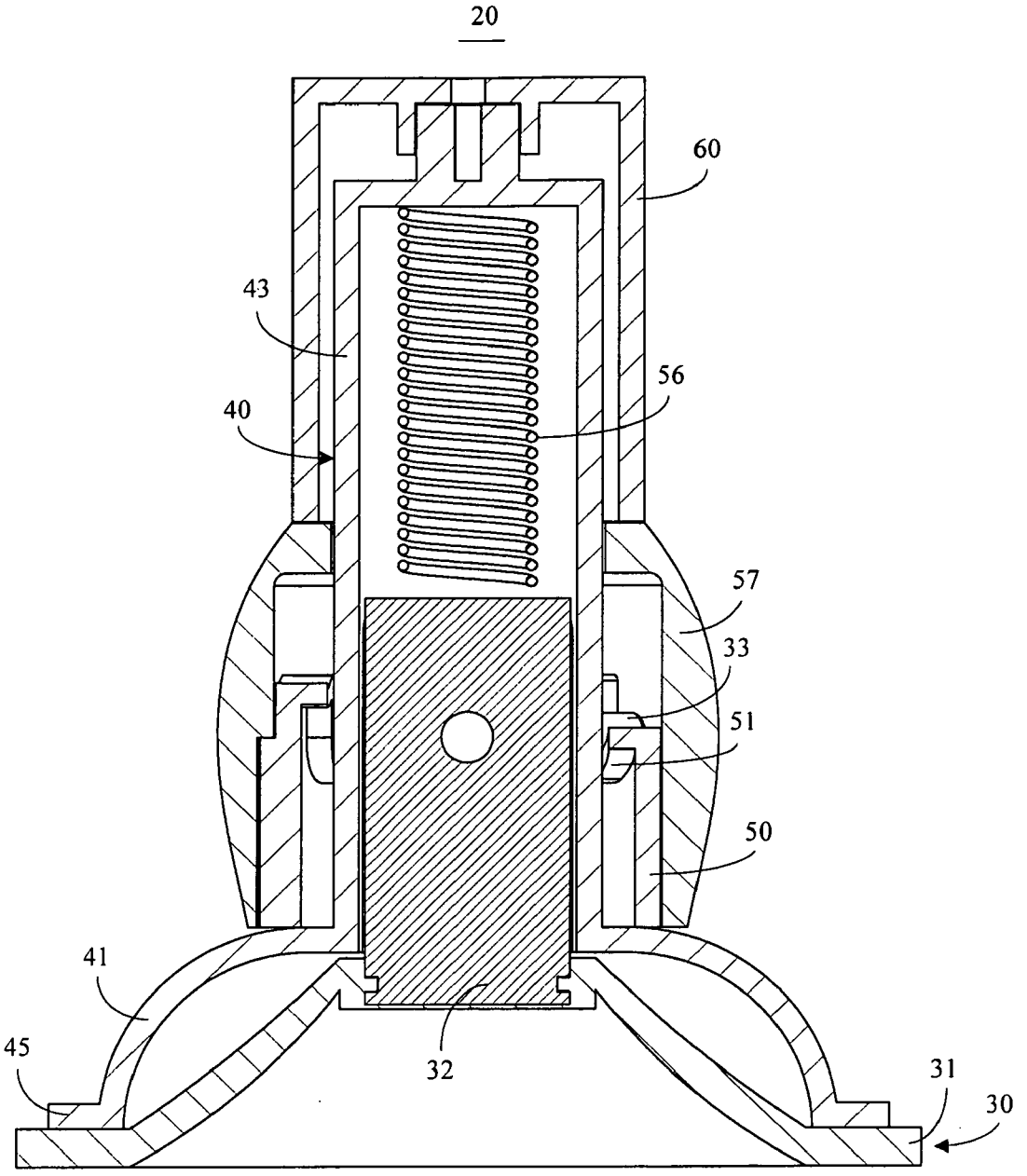


FIG. 7

RESIZABLE DEVICE STAND

BACKGROUND

[0001] 1. Technical Field

[0002] The present disclosure relates to a device stand.

[0003] 2. Description of Related Art

[0004] A motor vehicle may be used by different drivers who each have their mobile device they would like it to be supported by a stand in the vehicle, or a single driver may used different devices at different times. Because stands are usually adapted for a specific device type and size, many different stands must be available in the vehicle, which can be unsightly and waste space.

[0005] What is needed is a resizable stand adapted for supporting different types and sizes of devices.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] Many aspects of the embodiments can be better understood with reference to the following drawings. The components in the drawings are not necessarily drawn to scale, the emphasis instead being placed upon clearly illustrating the principles of the present disclosure. Moreover, in the drawings, like reference numerals designate corresponding parts throughout the several views.

[0007] FIG. **1** is a schematic, isometric view of a stand according to an exemplary embodiment, the stand including a suction base and a load-supporting module.

[0008] FIG. **2** is a exploded view of the load-supporting module of FIG. **1**.

[0009] FIG. **3** is a top plan view of the load-supporting module of FIG. **1**, with a top cover removed.

[0010] FIG. **4** is a schematic, isometric view of the suction base of FIG. **1**.

[0011] FIG. **5** is an exploded view of the suction base of FIG. **4**.

[0012] FIG. **6** is a section view of the suction base of FIG. **4** without a vacuum seal created.

[0013] FIG. **7** is a section view of the suction base of FIG. **4** with a vacuum seal created.

DETAILED DESCRIPTION

[0014] Referring to FIG. **1**, a stand **10** according to an exemplary embodiment is disclosed. The stand **10** includes a suction base **20** and a load-supporting module **70** mounted on a top of the suction base **20**.

[0015] Referring to FIGS. **2** and **3**, the load-supporting module **70** includes a housing **71**. The housing **71** is rectangular, and includes a top cover **72** and a bottom cover **75**. The top cover **72** defines a groove **73** therein. The housing **71** is configured to house a pair of clamps **80**, a gear **91**, a torsion spring **92**, and a locking structure **95**.

[0016] Each of the clamps **80** is L-shaped, and includes a vertical part **81** and a horizontal part **82**. The horizontal part **82** includes a rack gear **83**. The rack gear **83** defines a plurality of locking grooves **84** on outer sidewalls thereof. The horizontal part **82** is received by the groove **73** of the top cover **72**. Referring to FIG. **1**, the vertical parts **81** are positioned upwardly from the opposite sites of the housing **71**.

[0017] The gear **91** is rotatably mounted in the housing **71** and meshes with the rack gears **83**. The torsion spring **92** fits over a shaft of the gear **91**. The torsion spring **92** causes the gear **91** to rotate, and causes the clamps **80** meshing with the gear **91** to move away from each other.

[0018] The locking structure **95** is used to lock the clamps **80** to fix the clamps **80** at a position, or to unlock the clamps **80** to allow the clamps **80** to move.

[0019] In the exemplary embodiment, the locking structure **95** includes a locking pin **96**, a spring **97**, and an elastic piece **98**. The locking pin **96** is slidably mounted in the housing **71**. The spring **97** fits over the locking pin **96** and causes the locking pin **96** to engage with a locking groove **84** in one of the clamps **80** to lock the clamps **80**. The elastic piece **98** is rotatably mounted in the housing **71**. One end of the elastic piece **98** is fixed to the locking pin **96**. The other end of the elastic piece **98** stays in an opening **74** of the housing **71**. When the elastic piece **98** is pressed at the end in the opening **74**, the elastic piece **98** disengages the locking pin **96** from the locking groove **84**, thus unlocking the clamps **80**.

[0020] In other embodiments, the locking structure **95** can be a wedge slideably mounted in the housing **71**. The wedge can plug into gaps between teeth of the gear **91**, thus locking the gear **91** and the clamps **80**, or move away from the gear **91**, thus unlocking the gear **91** and the clamps **80**.

[0021] Referring also to FIG. **1**, when loading, a device (not shown) is placed on the housing **71**. The clamps **80** are pushed to move towards each other and to clamp the device. The locking structure **95** locks the clamps **80**. In this way, the device is fixed on the load-supporting module **70**. The distance between the clamps **80** can be varied for clamping different sized devices. When unloading a device, the elastic piece **98** of the locking structure **95** is pressed down and the clamps **80** are unlocked. The torsion spring **92** causes the clamps **80** to move away from each other. Then, the device can be unloaded from the load-supporting module **70**.

[0022] Referring to FIGS. **4**, **5** and **6**, the suction base **20** includes a suction piece **30**, a spring **56**, a fixing structure **40** for fitting over the suction piece **30**, pins **55**, a face cam **50**, a rotatable sleeve **57**, and a fixing cap **60**.

[0023] The suction piece **30** includes a membrane **31** and a shaft **32** extending upwardly from the membrane **31**.

[0024] The fixing structure **40** includes a shaft housing **41** and a sleeve **43** extending upwardly from the shaft housing **41**, and lower edge of the housing **41** forms a flange **45**. The sleeve **43** defines axial running slots **42**. An upper end of the sleeve **43** is closed.

[0025] The face cam **50** is ring-shaped and has a face **51** which includes grooves **53** defined therein. The face cam **50** further includes axial rails **52** projected from an outer circumferential surface thereof.

[0026] During assembly of the suction base **20**, first, the sleeve **43** receives the spring **56**, then is fit over the shaft **32**. The spring **56** is positioned between the upper end of the sleeve **43** and the shaft **32**. Second, the face cam **50** is fit over the sleeve **43**. One end of each of the pins **55** engages with the face **51** of the face cam **55**. The other end of each of the pins **55** extends through the running slots **42** defined in the sleeve **43** and enters into holes **33** defined in the shaft **32**. Third, the rotatable sleeve **57** fits over the face cam **50**, and grooves (not shown) defined in an inner surface of the rotatable sleeve **57** engage with the rails **52** of the face cam **50**, thus to connect the rotatable sleeve **57** to the face cam **50**. Finally, the fixing cap **60** is fixed to the upper end of the sleeve **43** by screws (not shown).

[0027] Referring to FIGS. **6** and **7**, in use, the suction base **20** is put on a flat surface, so as to ensure the membrane **31** is flush against the surface, and the pins **55** are initially at a lowest position on the face **51**. Then, the rotatable sleeve **57** is

grasped and rotated, while holding the base **20** against the surface. During rotation of the sleeve **57**, which causes the face cam **50** to rotate, the pins **55** are driven along the face **51** until engaging in the grooves **53**. Meanwhile, the shaft **32**, driven to move up by the pins **55**, pulls only a central portion of the membrane **31** away from the contacting surface, as the edge of the membrane **31** is limited by the flange **45** of the shaft housing **41**, thereby forming a vacuum seal between the membrane **31** and the contact surface. In this way, the suction base **20** is fixed on the surface.

[0028] The suction base **20** can be easily removed from the flat surface by rotating the rotatable sleeve **57**. In this way, the membrane **31** is driven to move down, the vacuum between the membrane **31** and the surface of the flat surface is eliminated, and the suction base **20** is no longer fixed on the flat surface.

[0029] Moreover, it is to be understood that the invention may be embodied in other forms without departing from the spirit thereof. Thus, the present examples and embodiments are to be considered in all respects as illustrative and not restrictive, and the invention is not to be limited to the details given herein.

What is claimed is:

1. A resizable device stand comprising:

a suction base; and

a load-supporting module mounted on the suction base, the load-supporting module comprising:

a housing defining a groove therein;

a pair of clamps, each clamp comprising a vertical part and a horizontal part, wherein the horizontal part comprises a rack gear, and is received by the groove, the vertical parts being positioned upwardly from opposite sides of the housing, respectively;

a gear rotatably mounted in the housing and meshing with the rack gears of the clamps, respectively; and

a locking structure for locking or unlocking the clamps.

2. The stand of claim **1**, further comprising a torsion spring configured for causing the gear to rotate, and thus for causing the clamps to move away from each other.

3. The stand of claim **2**, wherein the torsion spring is fit over a shaft of the gear.

4. The stand of claim **1**, wherein the rack gear define a plurality of locking grooves, the locking structure comprises a locking pin, a spring for causing the locking pin engage with the locking grooves of the rack gear, and an elastic piece configured for causing the locking pin to move away from the locking grooves when received a pressure.

5. The stand of claim **1**, wherein the locking structure is a wedge slideably mounted in the housing, the wedge is able to plug into gaps between teeth of the gear or move away from the gear.

6. The stand of claim **1**, wherein the clamps are L-shaped.

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